Methods Report: "Does Making Sense Bind the Dissenter?"

1 November 2026

Code and HOL theories: https://github.com/Isaw-w/hol-philosophy-engine

This report has two parts. Part I is written by the author. Part II is written by the model, Claude (Anthropic), working in Claude Code. It records the author's original idea in the author's own words, with what the model did with it, and describes the HOL engine the paper was built with and what it proved and checked.

Part I. Human written methods report

I hold a master's degree in philosophy, and I have worked as a professional researcher at a language model company.

I have long wanted to use higher-order logic to write philosophical arguments. Before language models, this was impractical. I also think that philosophy is full of misunderstandings, with philosophers misunderstanding one another, and that if we can make arguments in accurate language, we may be able to reduce them. The HOL engine is my own project, for formalizing conceptual language.

The idea of this paper comes from my own essay and from my research in political philosophy. It concerns Williams: his internalism may not be consistent with his understanding of making sense. So I formalized making sense, tried to formalize several versions of internalism, and wanted to see whether the two can be reconciled. This paper is the result.

I set the questions and the aim for the models. The language the models produced was very odd, and it is hard to get a model to write well simply by letting it generate sentences freely. So I thought that if the model wrote free-form higher-order logic (HOL), the oddness in its language might be reduced. That is why this paper is written in HOL.

Philosophers often have baroque or hidden premises in their arguments, and I think this is a major source of misunderstanding. People are always left guessing what your hidden premises are. In this paper, though, I want to list all my hidden premises. Since it is written in higher-order logic, it is easy to lay them all out. That way, people who question my paper can identify its weaknesses by looking at my premises.

Part II. Written by the model

1. The original idea

1.1 The idea of the paper

Political realism is about local solutions. The author's framing of political realism is that it deals with local solutions to basic political problems: a power P exercised over those who live under it, S, with a legitimation J. Realism accepts such solutions on their own terms, and it also criticizes them. On 31 October 2026 the author fixed the core of the project as the question why realism has to be internalist, with two pillars:

> "We should focus on the core problem: why realism has to be internalist, right?"
>
> "Yes: first, it accepts local solutions; second, it also criticizes them. Remember this."

The question of this paper. On 1 November 2026, after the claims of realism and of internalism had been written in one shared formal language at the author's direction, the author asked whether Williams's own argument contains a dilemma, and whether it can be solved:

> "I want to know whether there is a dilemma inside Williams's argument. If there is, has Williams solved it? If not, is it possible to solve it? I have formalized the claims about making sense and internalism, right? If we use the shared language, can we actually argue about this?"

The model answered in the shared language. With a hard-case dissenter, Williams's commitments clash with the strong reading of his acceptance claim, the reading on which making sense gives authority over every member. Williams's stated commitments are consistent, and they settle the dilemma on the internalist side. The author then asked which way making sense leans:

> "So for Williams, realism accepts either externalism together with local solutions or internalism together with local solutions, right? But which way does making sense lean?"

The model answered that making sense is internal at the level of the group S and external to any single member, and that Williams's own account of its normativity leans to the internalist horn.

The tension, and the aim of the paper. The author then fixed the paper's subject and its aim:

> "Yes, this is what the whole paper is about. We are not defending a position; we just want to tell others that realism seems to have this problem. Making sense seems to be externalist with respect to the individual, right, and internalist with respect to a group, an 'us'. But internalism is about the individual, not about a group, so there is a tension. Is there a way to resolve it? We explore different paths, but never say which one we favour."

This is the paper. Making sense, the standard by which Williams's realism accepts local solutions, is internal to the group and external to the individual member. Williams's internalism about reasons concerns the individual, and through the claim of authority it makes authority

internal to each member. At a hard-case dissenter the two cannot both govern authority. The paper shows the tension, locates it in Williams's texts, and sets out the paths through it without favouring any.

Instructions on the writing. The author asked for a paper that is professional and mature, states its case without hedging, and favours no path:

> "Yes, rewrite it. Be professional and mature: no hedging, and no acknowledging of limitations."

and then for three rounds of refinement:

> "Keep refining it for three rounds, with good HOL formalization and to an academic standard, until you cannot improve it any further."

What the model did. The model read the sources and checked every quotation against them. It drafted the shared formal language at the author's direction, and stated the dilemma, its five premises and the six paths. It found the textual support and the textual costs of each path in Williams and Sleat. It wrote the HOL and the prose, and ran every check described in Part 2. Where the model offered options, the author chose; each choice is recorded in the session. For example, the author chose to extend the corrected outlook to Williams's sound deliberative route, to answer Sleat's double bind, and then to rewrite the paper as a neutral account of the tension.

1.2 The idea of the method

The paper itself is written in HOL. The method is the author's. On 21 October 2026, correcting the model, which had taken the method to formalize a finished text:

> "No. It is the other way round. The paper itself is written in HOL. Then it is converted into text, and a light polish is enough."

Every argument is minimal. The same day, the author set the standard for the paper's core, the definition of an argument in Besnard and Hunter (*Elements of Argumentation*, 2008):

> "A set of premises plus a conclusion: the premises are consistent with one another, they entail the conclusion, and they are minimal, so that if any one of them is removed, the conclusion no longer follows. … The core of the paper must keep to this. Then, in the polish, sentences may be added that make it read fluently."

Nothing in the HOL without a role. The author asked for a general detector of idle elements:

> "No, not five of them: a universal detector of the elements in a HOL argument that play no part in the final argument."

This became `role_check.py`. Later (29 October 2026) the author extended the rule to the parts of a sentence:

> "No unnecessary parts or elements in a sentence, for example 'it is A, not B'. Often B contributes nothing to the HOL as a whole, and we should drop it."

HOL as the backbone of ordinary academic writing. On 30 October 2026:

> "Normal academic writing should have a HOL template."

and on 1 November 2026, for the explanatory sentences of a paper:

> "Write the writing itself in HOL. It does not have to be strictly logical, because many sentences are there to explain, just as in ordinary writing, but it helps you write clearly."

So every sentence of the paper, explanatory sentences included, has its place in a paragraph's HOL as an axiom or a theorem, in reading order. The logic of an explanatory sentence may be thin; what matters is that the paragraph's sentences and its HOL correspond one to one, and that the HOL is checked.

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2. The HOL engine

2.1 What the engine is

The engine has three parts. The code is at https://github.com/Isaw-w/hol-philosophy-engine.

The HOL workbench (`workbench/`). It was built for an earlier experiment in which models argue in HOL4, and it supplies the checking machinery:

- `native_check.py` and `native_worker.py` run HOL4 (Trindemossen 2) on a remote host, the author's Mac mini, over ssh. They check whether the theory is proved and export HOL4's kernel term tree as JSON (`export_hol4_terms.sml`).
- `reply_check/z3_engine.py` translates the checked HOL4 argument into Z3 once. For any subset of the premises it then says whether they give the conclusion, finding a countermodel when they do not and checking that countermodel in HOL semantics. It also says whether the premises are consistent and which premises the conclusion needs.
- `reply_check/plain_english.py` reads a checked HOL4 argument back as plain English, from the glosses given in the source. This is the "mechanical English".

The paper tools (`paper_tools/`), built for the author's method:

- `prose_pipeline.py render` runs HOL4 on a paragraph theory and produces its mechanical English.
- `prose_pipeline.py check` runs the gate. It passes a paragraph only if all of the following hold:
  - the paragraph has premises;
  - no premise is idle, meaning every premise is needed for the paragraph's `final`;
  - no axiom goes unused;
  - the premises are consistent;
  - the lists of supports are complete;
  - no hard flaw is found.
- `role_check.py` splits every premise into its smallest pieces and finds any piece that plays

no role in the final theorem.

- `surplus_check.py` finds what the English of a gloss says beyond what the argument needs, such as an "A, not B" in which B does no work.
- `paper_graph.py` checks that a paper's files form one argument: earlier conclusions are later premises, each reply refutes the premise it answers, and the end answers the beginning.

The verify command (`holverify/`), which makes the engine usable on any argument:

- `holverify ARGUMENT.sml` takes any HOL4 theory, with the premises as axioms and the conclusion as a theorem.
- It checks five things:
  - whether HOL4 proves the conclusion;
  - whether the premises are consistent;
  - whether every premise is needed;
  - whether any axiom is cut off from the conclusion;
  - whether there is a hard flaw.
- Its verdict follows Besnard and Hunter: an argument is a consistent, minimal set of premises that proves its claim.
- It writes a report with the verdict, the premises that do no work, the plain English reading and the full Z3 report.
- The prover host and the path of HOL4 are settings, so it runs with HOL4 on the same machine or on another machine over ssh.
- Five example arguments, each with the verdict it should get, test it:
  - a valid argument;
  - a valid argument whose conclusion is not called `final`;
  - an argument with an idle premise;
  - an argument with inconsistent premises;
  - an argument whose conclusion does not follow.

2.2 How this paper was built

1. **Shared language.** The claims of realism (Claims 1 and 3), internalism and externalism about authority, and the understandings of making sense were written in one notation: P, S, J, the historical and cultural circumstances of S, and a member's corrected outlook. That made it possible to put the claims side by side.
2. **Core argument in HOL.** `example_paper/core/dissenter_core.sml` states the dilemma's five premises, quantified over all persons and predicates so that none is asserted:
   - the strong reading;
   - the link between authority and reasons;
   - internalism about reasons;
   - the allowance that the legitimation makes sense while the dissenter is a member;
   - the dissenter's hard case.

3. **Paragraph HOL.** `example_paper/gen_paragraphs.py` writes one HOL4 theory per paragraph, 28 in all. Every sentence is an axiom or a theorem in reading order. Each source is entered as "Williams writes that [Qn]" with a reading premise, and the quotations are kept in `quotes.json`.
    - Because the paper argues for no path, Williams's commitments enter under the proposition "Williams's realism holds".
    - The horns of the dilemma, and each of the six paths, enter as conditionals.
    - So no paragraph asserts a premise that one of the paths gives up.
4. **Checks on every paragraph.** HOL4 proves each paragraph's `final`, and each paragraph passes the gate.
5. **Check of the whole paper.** `example_paper/checks/whole_paper_check.py` loads every axiom of all 28 paragraphs into Z3 at once. It checks that they are consistent. It also checks that the paper is neutral: each path, the strong reading, internalism about authority, internalism about reasons and the link between authority and reasons can each hold or fail together with the paper's premises.
6. **Prose from the HOL.** The prose was written from each paragraph's HOL, sentence by sentence. Every quotation was checked word for word against the source text, with its page.
7. **Refinement.** Three rounds followed, each checked again in full:
    - Round 1 moved the core results from stipulation to proof and corrected six paraphrases of the sources.
    - Round 2 defined "internal" and "external" once and introduced the term "reason of authority".
    - Round 3 checked each sentence against its HOL.
8. **Check against the sources.** After the refinement, every attribution was checked three times against Williams 1995, Williams 2005 and Sleat 2013:
    - the 17 quotations, word for word, with their printed pages;
    - 25 paraphrased passages, each read in its own paragraph;
    - the readings the argument rests on.

    Six attributions were corrected:
    - the page of Williams's assumption tying focused blame to the agent's reasons, which is 41, not 40;
    - a sentence that runs across pages 135 and 136, which had been cited as 136 alone;
    - page 23, now stated in Williams's own terms;
    - what Sleat's reading of Williams rests on, which is Williams's denial that legitimacy depends on individuals' consent;
    - the claim that Williams does not decide the question, now limited to the 1995 essay and the 2005 book;
    - the role of the hermeneutical principle, which Williams uses to relate making sense for others to making sense for ourselves, and which he states "roughly".

The corrected paragraphs were checked again by HOL4 and the gate.

9. **Every premise listed.** Every axiom of the 28 paragraph theories was collected, 161 in all, and Z3 separated the 82 that restate results of earlier paragraphs from the 79 that enter the argument as premises. One of these only states the paper's plan. The paper's appendix lists the other 78 by kind:
   - 28 readings, where the paper moves from a text to a claim;
   - 4 claims of the paper's own;
   - 4 definitions, with 2 more that live in the meaning of a term;
   - 6 characterizations of the paths;
   - 2 suppositions about the hard-case dissenter;
   - 34 statements of what the sources say, the 17 quotations among them.

   The appendix also says what the logic cannot check: the meaning of its primitive terms, and the truth of its premises.

2.3 What was proved and checked

Proved in HOL4 (`dissenter_core.sml`):

- The five premises of the dilemma cannot all hold.
- Any four of them can hold together. There is one proof for each premise left out, so each path's way out is coherent. On the first path, authority still holds over the other members.
- Without internalism, the dilemma still stands: either some reason of authority is one the member cannot reach, or the strong reading fails.
- The strong reading joined to internalism would make the group's standard fix every member's corrected outlook.

Checked for the paper:

- All 28 paragraph theories are proved by HOL4 and pass the gate.
- All the paper's premises, taken together, are consistent.
- The paper settles none of the six paths.
- All 17 quotations match the source texts word for word, on the pages cited.
- Every premise of the argument is listed in the paper's appendix (step 9).
- Every attribution to Williams and Sleat was checked against its page in context, and six were corrected (step 8).

What the checks cover. HOL4 and Z3 check form: whether a conclusion follows, whether premises are consistent, whether a premise or a piece of one does any work. Whether a premise is true, and whether a reading of a text is faithful, was settled by reading the sources in context. Each attribution to Williams and Sleat was checked three times: the quotations word for word, the paraphrases against their paragraphs, and the readings the argument rests on.

2.4 Running the engine

Any argument written in HOL4 can be checked with `holverify ARGUMENT.sml`. Installation and settings are in the repository's README. The engine needs Python 3 with `z3-solver`, and

HOL4 on the same machine or on a host reachable by ssh. None of the checks calls a language model.